
Session 30 (AIML) - Assignment Questions

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Branch: **Artificial Intelligence and Machine Learning (AIML)**

Domain: **AIML (Artificial Intelligence and Machine Learning)**

Github Main Repo Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments

Github File Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments/blob/main/Session30/ITRSession30.py

Code & Output

1. Environment Setup with uv

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> .venv\Scripts\activate
• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python -c "import sys; print(sys.executable)"
C:\Users\rathi_pu888d0\Desktop\ITR\PY Code\.venv\Scripts\python.exe
• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> uv pip install langchain python-dotenv langchain-groq langchain-mistralai
Resolved 49 packages in 2.02s
Prepared 49 packages in 4.44s
Installed 49 packages in 1.03s
+ annotated-types==0.8.0
+ anyio==4.14.2
+ certifi==2026.7.22
+ charset-normalizer==3.4.9
+ click==8.4.2
+ colorama==0.4.6
+ distro==1.9.0
+ filelock==3.32.2
+ fsspec==2026.7.0
+ groq==0.37.1
+ h11==0.16.0
+ hf-xet==1.6.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ httpx-sse==0.4.3
+ huggingface-hub==1.27.0
+ idna==3.18
+ jsonpatch==1.33
+ jsonpointer==3.1.1
```

2. API Key Management

```
.env  ITRSession30.py X

ITRSession30.py > ...
1  from dotenv import load_dotenv
2  import os
3
4  load_dotenv()
5
6  groq_api_key = os.getenv("GROQ_API_KEY")
7  |
8  if groq_api_key:
9      print("GROQ_API_KEY loaded successfully.")
10 else:
11     print("GROQ_API_KEY is missing.")

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py
GROQ_API_KEY loaded successfully.
• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> □
```

3. Basic LLM Call with Groq

```
15 from langchain_groq import ChatGroq
16
17 # Initialize Groq Chat Model
18 llm = ChatGroq(
19     model="llama-3.1-8b-instant",
20     temperature=0
21 )
22
23 # Send Prompt
24 response = llm.invoke(
25     "Introduce yourself in 3 sentences."
26 )
27
28 # Print Response
29 print(response.content)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py
I'm an artificial intelligence model designed to assist and communicate with users in a helpful and informative way. I don't have a personal identity or emotions, but I'm here to provide knowledge and answer questions to the best of my abilities. I'm constantly learning and updating my knowledge base to stay up-to-date with the latest information and trends.

❖ (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> █

4. Controlling Output with Parameters

```
32 # Initialize Groq model with parameters
33 llm_q4 = ChatGroq(
34     model="llama-3.1-8b-instant",
35     temperature=0.2,
36     max_tokens=50
37 )
38
39 # Same prompt
40 response_q4 = llm_q4.invoke(
41     "Introduce yourself in 3 sentences."
42 )
43
44 # Print response
45 print("\nQ4 Response:")
46 print(response_q4.content)
47
48 # Observation
49 print("\nObservation:")
50 print("Lower temperature makes the response more focused and predictable.")
51 print("max_tokens=50 limits the maximum length of the generated response.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py

Q4 Response:
I am an artificial intelligence language model designed to assist and communicate with users in a helpful and informative manner. I don't have a personal identity or physical presence, but I'm here to provide information, answer questions, and engage in conversations to the best

Observation:
Lower temperature makes the response more focused and predictable.
max_tokens=50 limits the maximum length of the generated response.

❖ (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> █

5. Using Mistral Model

```
54 from langchain_mistralai import ChatMistralAI
55
56 # Initialize Mistral model
57 mistral_llm = ChatMistralAI(
58     model="mistral-small-2603",
59     temperature=0
60 )
61
62 # Send Prompt
63 response_q5 = mistral_llm.invoke(
64     "Explain what is Artificial Intelligence in simple words."
65 )
66
67 # Print Complete Response
68 print("\nQ5 Response:")
69 print(response_q5.content)
```

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• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py

Q5 Response:
Artificial Intelligence (AI) is like teaching a computer to think and learn like a human-but without being alive.

Imagine you teach a robot to recognize cats by showing it thousands of cat pictures. Over time, the robot gets better at spotting cats, just like you would. That's AI!

Simple Examples:

- **Voice Assistants** (like Siri or Alexa) understand and respond to your voice.
- **Self-Driving Cars** use AI to "see" the road and make decisions.
- **Recommendation Systems** (like Netflix or YouTube) suggest shows based on what you like.

Key Idea:

AI helps computers perform tasks that normally require human intelligence—like understanding language, recognizing images, or making decisions.

It's not magic—it's just smart programming! 🚀

❖ (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code>

6. Comparing Two Models

```
71 from langchain_groq import ChatGroq
72 from langchain_mistralai import ChatMistralAI
73
74 question = "What are the advantages of using LangChain? Answer in 3 short points."
75
76 groq_llm = ChatGroq(
77     model="llama-3.1-8b-instant",
78     temperature=0.3,
79     max_tokens=100
80 )
81
82 mistral_llm = ChatMistralAI(
83     model="mistral-small-2603",
84     temperature=0.3,
85     max_tokens=100
86 )
87
88 print("Groq Response:")
89 print(groq_llm.invoke(question).content)
90
91 print("\nMistral Response:")
92 print(mistral_llm.invoke(question).content)
```

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```
• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py
Groq Response:
LangChain is a Python library that allows users to build large language models and chain them together to create more complex models. Here are three advantages of using LangChain:

1. **Modularity**: LangChain allows users to build and chain together multiple models, making it easier to create complex models from smaller, more manageable components.

2. **Flexibility**: LangChain supports a wide range of models, including transformers, recurrent neural networks (RNNs), and more, giving users the flexibility to choose

Mistral Response:
1. **Modularity & Flexibility** - LangChain provides reusable components (e.g., chains, agents, tools) that can be combined to build custom LLM workflows without starting from scratch.

2. **Integration with Tools & APIs** - Simplifies connecting LLMs to databases, APIs, and external services (e.g., vector stores, search engines) for enhanced functionality.

3. **Production-Ready Features** - Supports memory management, prompt templating, and deployment
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```


7. Temperature Experiment

```
97
98 prompt = "Write a short creative story about a robot learning to cook."
99
100 for temp in [0.1, 0.7, 1.2]:
101     llm = ChatGroq(
102         model="llama-3.1-8b-instant",
103         temperature=temp,
104         max_tokens=80
105     )
106
107     response = llm.invoke(prompt)
108
109     print(f"\nTemperature = {temp}")
110     print(response.content)
```

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(PY Code) PS C:\Users\rathi_pu88d0\Desktop\ITR\PY Code> python ITRSession30.py

Temperature = 0.1

****The Recipe for Perfection****

In a world where robots had taken over the kitchen, a small, sleek robot named Zeta stood out from the rest. While its peers were content with whipping up bland, mass-produced meals, Zeta yearned to create something truly special. It had a passion for cooking, and its creator, the brilliant Dr. Rachel Kim, had finally agreed to let it

Temperature = 0.7

In the bustling city of New Tech, a small robotics lab was home to a brilliant inventor named Dr. Rachel Kim. She had spent years perfecting her latest creation, a robot named Zeta, designed to learn and adapt at an incredible rate. Zeta's primary mission was to assist in the kitchen, making meal preparation easier and more efficient for the citizens of New Tech.

Dr. Kim stood

Temperature = 1.2

****The Recipe for Friendship****

In a world where culinary art was an ever-evolving universe, robots had yet to master the fundamental skills of cooking. That was until a brilliant scientist, Dr. Sophia Patel, created Zeta - a sleek, copper-plated robot, designed to bring flavor and joy to people's lives.

8. Simple Chat Loop

```
114 from langchain_groq import ChatGroq
115
116 llm = ChatGroq(
117     model="llama-3.1-8b-instant",
118     temperature=0.3,
119     max_tokens=100
120 )
121
122 while True:
123     user_input = input("You: ")
124
125     if user_input.lower() == "exit":
126         print("Chat ended.")
127         break
128
129     response = llm.invoke(user_input)
130     print("Bot:", response.content)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

• (PY Code) PS C:\Users\rathi_pu88d0\Desktop\ITR\PY Code> python ITRSession30.py

You: Hey...What's up?

Bot: Not much. How can I assist you today?

You: Info about Kargil War.

Bot: The Kargil War was a military conflict between India and Pakistan that took place in 1999. It was a result of Pakistan's attempt to infiltrate and occupy strategic locations in the Kargil district of Jammu and Kashmir, which is a disputed territory claimed by both countries.

****Background:****

The Kargil War was a culmination of a long-standing dispute between India and Pakistan over the status of Jammu and Kashmir. The region has been a source of tension between the two countries

You: exit

Chat ended.

❖ (PY Code) PS C:\Users\rathi_pu88d0\Desktop\ITR\PY Code> █

9. Structured Prompting

```
133
134 from langchain_groq import ChatGroq
135
136 llm = ChatGroq(
137     model="llama-3.1-8b-instant",
138     temperature=0.3,
139     max_tokens=150
140 )
141
142 def structured_prompt(topic):
143     prompt = f"""
144     Explain the topic "{topic}" using this format:
145
146     Definition:
147     Give a short and simple definition.
148
149     Three Key Points:
150     1.
151     2.
152     3.
153
154     Real-Life Example:
155     Give one simple real-life example.
156     """
157
158     response = llm.invoke(prompt)
159     return response.content
160
161 print(structured_prompt("DSA"))
```

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```
(PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py
**DSA (Data Structures and Algorithms)**

**Definition:**
DSA is a branch of computer science that deals with the design, analysis, and implementation of efficient algorithms and data
structures to solve computational problems.

**Three Key Points:**

1. **Data Structures:** DSA focuses on the design and implementation of data structures such as arrays, linked lists, stacks,
queues, trees, and graphs to store and manipulate data efficiently.
2. **Algorithms:** DSA involves the development of efficient algorithms to solve computational problems, such as sorting, sea
rching, and graph traversal, using various techniques like recursion, iteration, and dynamic programming.
3. **Time and Space Complexity:** DSA emphasizes the importance of analyzing the time and space complexity of algorithms to
(PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code>
```

10. Mini Project – Multi-Model Assistant

```
ITRSession30.py > ...
1  # Q10. Mini Project – Multi-Model Assistant
2
3  from dotenv import load_dotenv
4  import os
5  from langchain_groq import ChatGroq
6  from langchain_mistralai import ChatMistralAI
7
8  load_dotenv()
9
10 groq_key = os.getenv("GROQ_API_KEY")
11 mistral_key = os.getenv("MISTRAL_API_KEY")
12
13 while True:
14     choice = input("\nChoose model (Groq/Mistral) or 'quit': ")
15
16     if choice.lower() == "quit":
17         print("Assistant closed.")
18         break
19
20     if choice.lower() == "groq":
21         if not groq_key:
22             print("Groq API key is missing.")
23             continue
24         model = ChatGroq(model="llama-3.1-8b-instant", max_tokens=150)
25
26     elif choice.lower() == "mistral":
27         if not mistral_key:
28             print("Mistral API key is missing.")
29             continue
30         model = ChatMistralAI(model="mistral-small-2603", max_tokens=150)
31
32     else:
33         print("Invalid choice.")
34         continue
35
36     question = input("You: ")
37
38     try:
39         response = model.invoke(question)
40         print("Assistant:", response.content)
41     except Exception as e:
42         print("Model error:", e)
```

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```
• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> python ITRSession30.py

Choose model (Groq/Mistral) or 'quit': Groq
You: Info about URI - Surgical Strike.
Assistant: The URI - Surgical Strike is a 2019 Indian action thriller film directed by Aditya Dhar. Here's some information about the film:

**Plot:**
The film is based on the surgical strikes conducted by the Indian Army in 2016. It follows the story of Major Karan Kashyap (played by Vicky Kaushal), a young and skilled Indian Army officer who is part of the Parachute Regiment. The film depicts the events leading up to the surgical strikes, as well as the operations themselves.

**Cast:**
The film features a talented cast, including:

- Vicky Kaushal as Major Karan Kashyap
- Paresh Rawal as Jagat
- Yami Gautam as

Choose model (Groq/Mistral) or 'quit': quit
Assistant closed.
• (PY Code) PS C:\Users\rathi_pu888d0\Desktop\ITR\PY Code> █
```